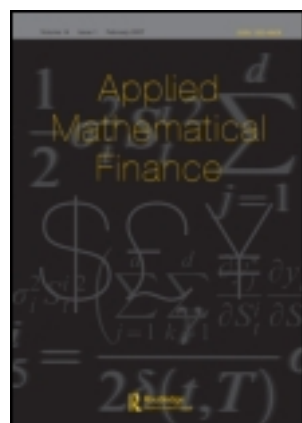


This article was downloaded by: [University of Chicago]

On: 18 February 2013, At: 10:10

Publisher: Routledge

Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Applied Mathematical Finance

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/ramf20>

Binomial models for option valuation - examining and improving convergence

Dietmar P. J. Leisen^a & Matthias Reimer^a

^a Department of Statistics, Rheinische Friedrich-Wilhelms-Universität Bonn, Adenauerallee 24-42, Bonn, D-53113, Germany

Version of record first published: 24 May 2006.

To cite this article: Dietmar P. J. Leisen & Matthias Reimer (1996): Binomial models for option valuation - examining and improving convergence, Applied Mathematical Finance, 3:4, 319-346

To link to this article: <http://dx.doi.org/10.1080/13504869600000015>

PLEASE SCROLL DOWN FOR ARTICLE

Full terms and conditions of use: <http://www.tandfonline.com/page/terms-and-conditions>

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden.

The publisher does not give any warranty express or implied or make any representation that the contents will be complete or accurate or up to date. The accuracy of any instructions, formulae, and drug doses should be independently verified with primary sources. The publisher shall not be liable for any loss, actions, claims, proceedings, demand, or costs or damages whatsoever or howsoever caused arising directly or indirectly in connection with or arising out of the use of this material.

Binomial models for option valuation – examining and improving convergence

DIETMAR P. J. LEISEN and MATTHIAS REIMER

Department of Statistics, Rheinische Friedrich-Wilhelms-Universität Bonn, Adenauerallee 24-42, D-53113 Bonn, Germany

Received January 1996

Binomial models, which describe the asset price dynamics of the continuous-time model in the limit, serve for approximate valuation of options, especially where formulas cannot be derived analytically due to properties of the considered option type. To evaluate results, one inevitably must understand the convergence properties. In the literature we find various contributions proving convergence of option prices. We examine convergence behaviour and convergence speed. Unfortunately, even in the case of European call options, distorted results occur when calculating prices along the iteration of tree refinements. These convergence patterns are examined and order of convergence one is proven for the Cox–Ross–Rubinstein model as well as for two alternative tree parameter selections from the literature. Furthermore, we define new binomial models, where the calculated option prices converge smoothly to the Black–Scholes solution, and we achieve order of convergence two with much smaller initial error. Notably, only the formulas to determine the up- and down-factors change. Finally, following a recent approach from the literature, all tree approaches are compared with respect to speed and accuracy, calculating the relative root-mean-squared error of approximate option values for a sample of randomly selected parameters across a set of refinements. Here, on average, the same degree of accuracy is achieved 1400 times faster with the new binomial models. We also give some insights into the peculiarities in the valuation of the American put option. Inspecting the numerical results, the approximation of American-type options with the new models exhibits order of convergence one, but with a smaller initial error than with previously existing binomial models, giving the same accuracy on average *ten-times faster than previous binomial methods*.

Keywords: binomial model, option valuation, order of convergence, convergence pattern

1. Introduction

With arbitrage pricing theory, the present value of any derivative security is obtained by calculating the initial cost of a payoff-replicating dynamic portfolio strategy in the underlying asset and the savings account. Commonly, it is assumed that the underlying asset follows a stochastic log-normal diffusion process. Consequently, in this set-up the replicating portfolio is adjusted continuously. Alternatively, in a simplified approach, asset price changes are decomposed into a sequence of Bernoulli steps implying a time- and state-discrete replicating strategy.

We are grateful to Hans Föllmer, Stefan Look, Dieter Sondermann *et al.* for comments and assistance. Financial support from the Deutsche Forschungsgemeinschaft, Sonderforschungsbereich 303 and from the Graduiertenkolleg, both at Rheinische Friedrich-Wilhelms-Universität Bonn, is gratefully acknowledged.

For practical applications however, a binomial tree of only fixed length approximates the continuous set of trading occurrences and asset prices by always covering some finite range of asset prices with a discrete grid structure. Using binomial trees for option pricing, one inevitably must examine the approximation quality by careful consideration of the approximation result with changing tree refinements.

Investigations reveal that the acquired degree of precision in binomally computed option prices in comparison with the continuous-time limit varies with the refinement of the binomial trees. The option prices asymmetrically oscillate with changing amplitude around the Black–Scholes solution for a European call option.

In general, an approximation method should have a convergence speed as great as possible. Convergence speed is measured here by the order of convergence in price approximations. This paper goes beyond the findings of the existing literature in several ways. First, for several existing lattice approaches the order of convergence is derived. In the literature, there are numerous contributions with limit proofs to European type options. Astonishingly however, the convergence speed of binomially computed option prices has, so far, rarely been examined technically. Here, we present a theorem which characterizes the order of convergence as the reduced by one highest order given for the first, second and third moments and a pseudo-moment. This allows for a comparison of methods according to well measurable and quantifiable criteria.

Secondly, the presentation of models with order of convergence two and a smooth convergence pattern follows. We exploit findings of the mathematical literature on normal approximations to the binomial function. The approach is general with respect to the choice of inverted normal approximations. In particular, there is the Peizer and Pratt (1968) method which consists of a convolution of several methods.

We also give some insights into the peculiarities in the valuation of the American put option. We transfer our findings about irregularities occurring with the pricing of European-type options.

Finally, we give some numerical examples to underline the strength of the new approximation. A method recently presented by Broadie and Detemple (1996) follows, where option prices are computed for a large sample of random parameters and then the relative standard deviation to the true solution is calculated and compared to computation time with increasing refinement. It is shown graphically that previous models lag drastically behind with respect to accuracy.

Using the same sample of parameters it is shown that our tree models improve the approximation of American put option prices. Admittedly, we do not obtain order of convergence two, but an improvement results because the European put price is the major component in the price of the American put option price.

A binomial option pricing model was first developed simultaneously by Cox, Ross and Rubinstein (1979) (CRR) and Rendleman and Bartter (1979). CRR presented the fundamental economic principles of option pricing by arbitrage considerations in the most simplest manner. By application of a Central Limit Theorem (CLT) they proved that their model merges into the Black and Scholes model when the time steps between successive trading instances approach zero.

The first group of articles focuses on lattice approaches with supposedly improved approximation properties. Jarrow and Rudd (1983) (JR) constructed a binomial model where the first two moments of the discrete and continuous time log return processes match. Regardless of the given tree refinement, the martingale measure is equal to one half here. Boyle (1988) constructed a

trinomial lattice, which is fixed up to some arbitrary parameter λ , which is determined heuristically. Although this model lacks a universal solution, he realizes that there are possibilities to improve lattice approaches by an ingenious choice of parameters.

Omberg (1988) deduced a whole family of lattice trees using the technique of Gauss–Hermite quadrature as a solution to the backward recursive integration problem. Unfortunately, trees with four or more vertices do not recombine properly and interpolation methods must be applied to keep a tractable grid of asset prices. Notably, he recognizes that even with a 20th order Gauss–Hermite jump process, the location of the exercise price within the tree structure may cause trouble.

Tian (1993) proposed binomial and trinomial models. The model parameters are derived as unique solutions of equation systems where the first two and higher moments of the continuous and discrete asset-distributions are matched. Unfortunately, this interesting contribution lacks rigorous mathematical arguments to support its ideas.

The second group of articles focuses on the valuation of exotic options. Numerous adjustments have been introduced to apply lattice approaches to various types of options. Already, Cox and Rubinstein (1985) have presented an adjustment for the valuation of Down-and-out calls. Hull and White (1993) modified the original CRR-model for the pricing of path-dependent exotic options by linear or quadratic interpolation. Recently, Cheuk and Vorst (1994) presented a model where the pay-off of Lookback options itself is modelled in a lattice, thus resolving the path dependency. Boyle *et al.* (1989) developed an n -dimensional extension of the lattice binomial method for valuing multivariate contingent claims. Madan *et al.* (1989) generalized the CRR-model to the multinomial case and investigated limits. He (1990) generalized the CRR-model to a multivariate multinomial model and in that context he showed convergence for prices and hedge strategies.

The last group of articles focused on approaches to construct lattices for more complex specific stochastic processes. Frequently the convergence issue is addressed. Nelson and Ramaswamy (1990) gave a general treatment including convergence proofs to the construction of simple binomial processes for various asset price dynamics. Amin (1991) developed a class of discrete, path-independent models, including models with time-varying volatility functions and multiple state variables. Li (1992) explored the potential of transformation and other schemes to construct simple binomial process, e.g. for state varying volatility functions. Amin and Khanna (1994) proved convergence for American options under a fairly general set of conditions.

2. On discrete and continuous models

Black and Scholes assumed that trading at financial markets proceeds continuously in time. Asset-price dynamics are described by

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (1)$$

where r is the instantaneous expected return of the underlying asset $S = (S_t)_{t \geq 0}$ if the risk-neutrality argument of Harrison and Pliska (1981) is used immediately, σ^2 is the instantaneous variance of the return, and $(W_t)_{t \geq 0}$ is a standard Gauss–Wiener process on a suitable probability space $(\Omega, \mathcal{F}, P)$.

Within this model, for any fixed European call option, having strike price K and maturity date

T , a hedge portfolio can be constructed containing solely the underlying asset S and a savings account with riskless borrowing and lending at r , which perfectly replicates the value of this option¹ at each instant t of time.

It is this no-arbitrage condition that leads to the Black–Scholes differential equation for the unknown function c

$$\frac{\partial c}{\partial t} + rS \frac{\partial c}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 c}{\partial S^2} = rc \quad (2)$$

The solution to the differential equation with boundary condition $f: x \mapsto (x - K)^+$ as the pay-off function is given by the Black–Scholes option pricing formula

$$c(t, S) = S \cdot \mathcal{N}(d_1) - K \cdot e^{-r(T-t)} \mathcal{N}(d_2) \quad (3)$$

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma(T-t)^{1/2}}$$

where $\mathcal{N}(\cdot)$ is the cumulative standard normal distribution function. Due to Harrison and Pliska (1981) we know that this can be written as

$$c(t, S) := e^{-r(T-t)} E[f(S_T)] \quad (4)$$

Transferring this framework into the simplifying binomial structure induces several adjustments. Suppose there is given a second probability space $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{P})$. Given a prespecified number n of trading dates, we suppose that trading occurs only at equidistant spots of time $t_i^n \in \{0 = t_0^n, \dots, t_n^n = T\}$ with $t_{i+1}^n - t_i^n \Delta t_n = \frac{T}{n}$ ($i = 0, \dots, n-1$). The one-period returns $\bar{R}_{n,i}$ ($i = 1, \dots, n$) are modelled by two point iid binomial random variables

$$\bar{R}_{n,i} = \begin{cases} u_n & \text{with probability } p_n \\ d_n & \text{with complementary probability } 1 - p_n \equiv q_n \end{cases} \quad (5)$$

on $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{P})$. Thus, for all $k = 0, \dots, n$ the discrete asset price dynamics at time t_k^n are described by

$$\bar{S}_{n,k} = S_0 \prod_{i=1}^k \bar{R}_{n,i} \quad (6)$$

The specification of the one-period returns is a complete description of the discrete dynamics $\bar{S}_n$. We call such a finite sequence $\bar{R}_n = (\bar{R}_{n,i})_{i=1, \dots, n}$ a *lattice (tree)*.

The parameters u_n, d_n, p_n, n differ from lattice to lattice, but remain constant throughout a specific lattice. For given r, σ, S_0, T the specification of these parameters for each refinement n is performed by the so-called *lattice approach*.

Several different lattice approaches have been proposed. They all take into account that the risk-neutrality argument of Harrison and Pliska (1981) implies that the expected one-period return $\bar{E}[\bar{R}_{n,1}]$ must equal the one-period return of the riskless bond $r_n = \exp\{r\Delta t_n\}$. The remaining degrees of freedom are resolved in different ways as shown in Table 1. Within all these models,

¹ The value depends on time t and asset price S under consideration. We adopt the common notation $c(t, S)$ for this function.

Table 1. Remaining degrees of freedom are resolved in different ways in the considered lattice approaches of CRR, JR and Tian; thus there are alternative definitions of tree parameters.

$u_n = \exp\left\{\sigma\sqrt{\frac{T}{n}}\right\}$	$u_n = \exp\left\{\mu' \frac{T}{n} + \sigma\left(\frac{T}{n}\right)^{1/2}\right\}$	$u_n = \frac{r_n v_n}{2}(v_n + 1 + (v_n^2 + 2v_n - 3)^{1/2})$
$d_n = \exp\left\{-\sigma\sqrt{\frac{T}{n}}\right\}$	$d_n = \exp\left\{\mu' \frac{T}{n} - \sigma\left(\frac{T}{n}\right)^{1/2}\right\}$	$d_n = \frac{r_n v_n}{2}(v_n + 1 - (v_n^2 + 2v_n - 3)^{1/2})$
	$\mu' = r - \frac{1}{2}\sigma^2$	$r_n = \exp\left\{r \frac{T}{n}\right\}$
		$v_n = \exp\left\{\sigma^2 \frac{T}{n}\right\}$

a hedge portfolio can be constructed which perfectly replicates the value of a European call option at each discrete point of time t_i^n regardless of whether the asset price increases to $\bar{S}_{n,i+1} = u_n \bar{S}_{n,i}$ or decreases to $\bar{S}_{n,i+1} = d_n \bar{S}_{n,i}$. Nodewise pay-off replication with respect to up- and down-movements requires

$$\Delta_i u_n \bar{S}_{n,i} + r_n B_i = c_n(t_{i+1}^n, u_n \bar{S}_{n,i})$$

$$\Delta_i d_n \bar{S}_{n,i} + r_n B_i = c_n(t_{i+1}^n, d_n \bar{S}_{n,i})$$

to hold.

From this, the proportion Δ_i of the underlying asset and amount B_i of savings in the replicating portfolio can be derived, the value at t_i^n is interpretable as the discounted expected value of prices at t_{i+1}^n with martingale measure equal to $p_n = (r_n - d_n)/(u_n - d_n)$. Here, the recursive replication procedure can be subsumed for all discrete steps in a binomial formula, of which the first was presented by CRR

$$c_n(0 = t_0^n, S_0) = r_n^{-n} \bar{E}[f(\bar{S}_{n,n})|S_0] \tag{7}$$

$$= r_n^{-n} \sum_{j=0}^n \binom{n}{j} p_n^j (1 - p_n)^{n-j} [\bar{S}_{n,n} - K]^+ \tag{8}$$

and equivalently

$$c_n(0 = t_0^n, S_0) = S_0 \Phi[a; n, p'_n] - K r_n^{-n} \Phi[a; n, p_n] \tag{9}$$

where

$$p_n = \frac{r_n - d_n}{u_n - d_n} \quad p'_n = \frac{u_n}{r_n} \cdot p_n \quad a = \left[\frac{\ln(K/S_0) - n \ln d_n}{\ln u_n - \ln d_n} \right]$$

where $\Phi[\cdot]$ denotes the binomial distribution function.

Formulas (7)–(9) give an approximation of the Black–Scholes price (3). This price approximation is a consequence of the approximation of the process (1) and therefore it implicitly gives the option price dynamics through the backward replication argument above.

The lattice approach is very similar to explicit finite difference methods. Explicit finite difference methods start immediately from the Black–Scholes partial differential equation (2) by discretizing

it suitably, in order to get price approximations. Lattice approaches start with a particular description of the asset price dynamics. Explicit finite difference methods focus on the dynamics of the no-arbitrage portfolio, only implicitly regarding the asset price dynamics itself. For a detailed description of explicit finite difference methods and their connection to lattice approaches we refer the reader to Wilmott *et al.* (1993). Geske and Shastri (1985) compared the two approaches. They came to the conclusion that depending on the situation each has advantages over the other.

3. Examining convergence

All presented lattice approaches are constructed in such a way that $\bar{S}_{n,n}$ weakly converges to S_T . Taking the logarithms this can easily be seen as follows. Denoting the mean and variance of $\ln \bar{S}_{n,n}$ by $\hat{\mu} \cdot n$, $\hat{\sigma}^2 \cdot n$ and the mean and variance of $\ln S_T$ by μt , $\sigma^2 t$, the CLT together with the Ljapunov-condition states that the following conditions are sufficient to ensure the required convergence

$$\hat{\mu} n \xrightarrow{n} \mu t \quad (10)$$

$$\hat{\sigma}^2 n \xrightarrow{n} \sigma^2 t \quad (11)$$

$$\frac{\sum_{k=1}^n E[(\ln \bar{R}_{n,k} - \hat{\mu})^3]}{(\hat{\sigma}\sqrt{n})^3} \xrightarrow{n} 0 \quad (12)$$

Since, for all three models, these requirements are clearly fulfilled, weak convergence of the terminal distribution is fulfilled. Convergence of prices is a consequence of weak convergence of the respective distributions.

Our interest, however, focuses on convergence behaviour and convergence speed. These two topics we address here.

Figure 1 depicts a typical pattern resulting from plotting option prices obtained by different tree refinements. The straight horizontal line indicates the Black–Scholes solution. For the binomial calculations we plot the result for each refinement and draw a connecting line to emphasize the changing results. For all three models we find oscillations and wave patterns. Furthermore, intervals with decreasing errors are followed by intervals with again increasing errors².

To understand the approximation pattern one has to look at the sequence of discrete distribution functions. In Fig. 2 the vertical dashed lines mark the terminal nodes of a ten-step binomial tree. The smooth line gives the continuous density function along a continuous set of asset prices. The rectangles surrounding the terminal nodes form a histogram. However, contrary to this illustration, these probability masses are concentrated at the node-points, implying that splitting the probability mass with the exogenous fixed strike price occurs in whole rectangles only. Translating into a binomial distribution function we have a piecewise constant function with jumps at each terminal node.

² We did not depict calculations of an at-the-money option with the CRR-model, because the homogenous convergence pattern there represents a unique exception. Moreover, we omitted calculations for the JR and Tian model, since they are similar in appearance.

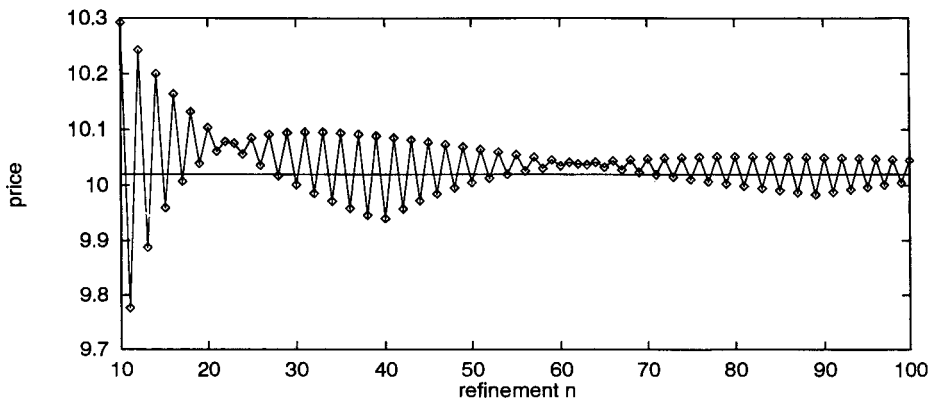


Fig. 1. Typical pattern resulting from option price calculations with binomial models: example with CRR model and the following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 100$.

However, the set of binomial asset prices moves with each iteration of the tree refinement in the way indicated in Fig. 3.

As the sequences of $(u_n)_n$ and $(d_n)_n$ converge to one the tree grid seems to migrate to the initial asset price. A strike location at points across these streamlines with changing distances to surrounding tree nodes results. As a result, the computed option prices oscillate and wavily converge to the Black–Scholes solution³.

Alternatively, in order to visualize that the fluctuating separation causes existing convergence patterns, take the Black–Scholes formula and adapt the strike price in the normal components to the implicit separation rule of the binomial models

$$d_{1,2} = \frac{\ln(S/\hat{K}) + (r \pm \frac{1}{2}\sigma^2)(T - t)}{\sigma(T - t)^{1/2}} \text{ where } \hat{K} = S_0 u^a d^{n-a}$$

The existing convergence patterns with all the considered binomial models with arbitrary parameter selections can be reproduced, displaced with respect to the distribution error only⁴.

Next, we will study convergence speed. Suppose we take one sequence of lattices in particular. Typically, discrete and continuous prices do not match. We find an error $e_n = |c(0, S_0) - c_n(0, S_0)|$. Due to the CLT we have $\lim_{n \rightarrow \infty} e_n = 0$, which means prices calculated by a sequence of lattices converge to its respective Black–Scholes price.

Due to the wavy patterns in the convergence behaviour, the adequate formal description of the approximation speed is to characterize convergence by use of a suitable upper bound for the error e_n . For this we use the mathematical concept of ‘order of convergence’. Restated for our specific case here, we make the following definition.

³ This property is sometimes called the ‘even-odd-problem’.
⁴ The distribution error can be quantified by adapting the Black–Scholes formula once more, now with a series expansion. Thus, binomial option prices can be reproduced by this twofold adaptation.

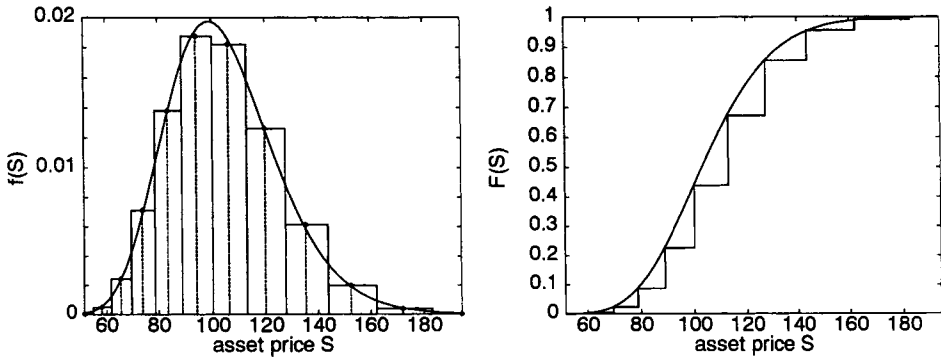


Fig. 2. A histogram depicting the connection to the continuous density function; separation of the probability mass only with whole rectangles gives a distribution function with steps; example drawn from a 10-step binomial tree.

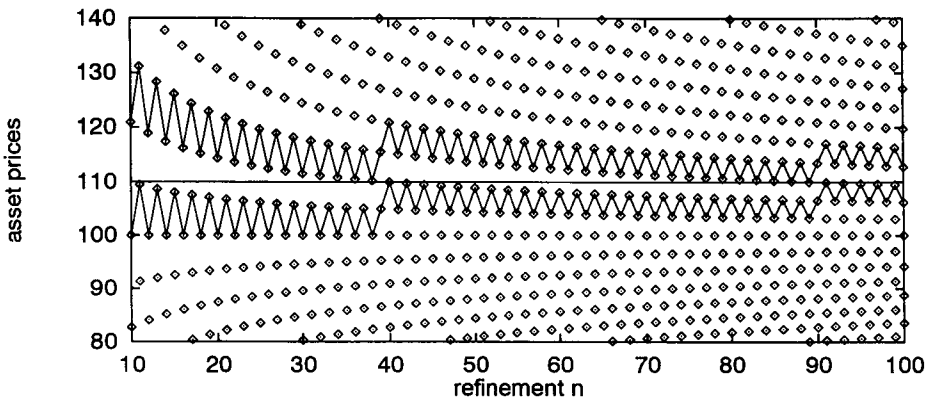


Fig. 3. Terminal asset price grids with iteration of the tree refinement and location of the strike price: example with CRR-model and the following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 100$.

Definition⁵

Let $f : x \mapsto (x - K)^+$ be a European call option. A sequence of lattices *converges with order* $\rho > 0$ if there exists a constant $\kappa > 0$ such that

$$\forall n \in \mathbb{N} : e_n \leq \frac{\kappa}{n^\rho} \quad (13)$$

⁵ All considerations are performed for European call options only. However due to put-call-parity the results immediately carry over to European put options as well.

A lattice approach converges with order $\rho > 0$ if for all S_0, K, r, σ, T the specified sequence of lattices converges with order $\rho > 0$.

In the following we will often denote this with $e_n = \mathcal{O}(\frac{1}{n^\rho})$. Please note that convergence of prices is implied by any order greater than 0. Higher order means ‘quicker’ convergence. Thus, the theoretical concept of order of convergence is not unique: a lattice approach with order ρ also has order $\tilde{\rho} \leq \rho$.

Order of convergence is very easy to observe in actual simulations: in the figures, we plot the error e_n against the refinement n on a log-log scale. Because $\log \kappa/n^\rho = \log \kappa - \rho \log n$ the bounding function κ/n^ρ becomes a straight line with slope equal $-\rho$ and shift κ . For example, in Fig. 4, order of convergence one is suggested. We could present many of these figures. All suggest order of convergence one. Our aim is to identify unique criteria which determine the order of convergence of a specific lattice approach under consideration in mathematical terms. As a starting point we make the following definition.

Definition

For a sequence of lattices $(\bar{R}_n)_{n \in \mathbb{N}}$ we call for all $n \in \mathbb{N}$

$m_n^1 := \bar{E}[\bar{R}_{n,1} - 1] - E[R_{n,1} - 1]$ (14)

$m_n^2 := \bar{E}[(\bar{R}_{n,1} - 1)^2] - E[(R_{n,1} - 1)^2]$ (15)

$m_n^3 := \bar{E}[(\bar{R}_{n,1} - 1)^3] - E[(R_{n,1} - 1)^3]$ (16)

moments and

$p_n := \bar{E}[(\ln \bar{R}_{n,1})(\bar{R}_{n,1} - 1)^3]$ (17)

pseudo-moment

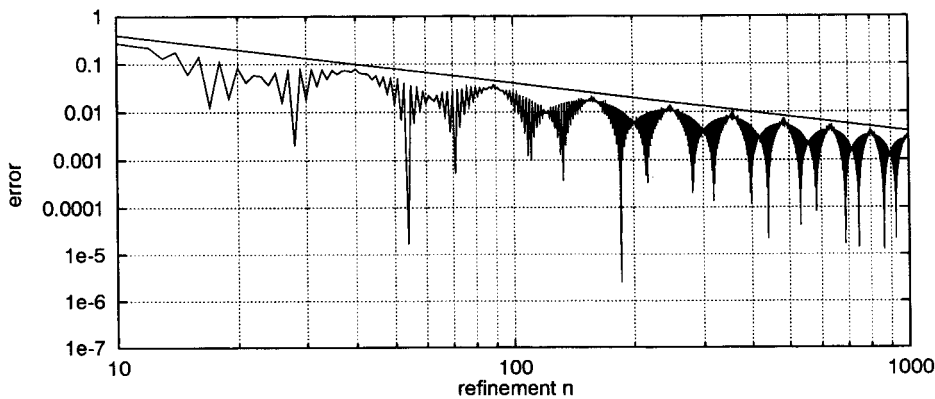


Fig. 4. Graphical representation and examination of the error bound; x-axis and y-axis with log-scale; example with CRR-model and the following selection of parameters: $S = 100, K = 110, T = 1, r = 0.05, \sigma = 0.3, n = 10, \dots, 1000$.

For any $n \in \mathbb{N}$ we denote by $R_n = (R_{n,i})_{i=1,\dots,n}$ the continuous return between times t_i^n and t_{i+1}^n . They are iid random variables on $(\Omega, \mathcal{F}, P)$ such that $S_{t_k^n} = S_0 \prod_{i=1}^k R_{n,i} \forall k = 0, \dots, n$.

Our moments are a generalization of the ordinary per period moments being the difference between the ordinary per period moments of the discrete and the continuous approach. It can be shown easily that (10) to (12) imply convergence of the moments $\mathbf{m}_n^1$, $\mathbf{m}_n^2$ and $\mathbf{m}_n^3$ to zero. This holds for every weakly converging lattice approach. In the following we are able to establish the result that the moments defined above are the determining factors for the order of convergence of prices in particular. Knowing that a lattice converges with order $\rho > 0$ clearly implies convergence of prices. However, convergence of prices does not give us any information about the order of convergence. Unfortunately, it turns out that a suitable convergence of moments is not sufficient to ensure order of convergence ρ alone. The proof of Theorem 1 requires the pseudo-moment. Its form is of a technical nature, as it results from our proof.

Theorem 1

Let $(\bar{R}_n)_{n \in \mathbb{N}}$ be a sequence of lattices and $\mathbf{m}_n^2$, $\mathbf{m}_n^3$, $\mathbf{p}_n$, its respective (pseudo-) moments. The order of convergence is the smallest order contained in $\mathbf{m}_n^2$, $\mathbf{m}_n^3$, and $\mathbf{p}_n$ reduced by 1, but not smaller than 1, that is

$$\exists \kappa(S_0, K, r, \sigma, T) : e_n \leq \kappa \left\{ n(\mathbf{m}_n^2 + \mathbf{m}_n^3 + \mathbf{p}_n) + \frac{1}{n} \right\} \quad (18)$$

Proof⁶

Denote by $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{P})$ the product space of $(\Omega, \mathcal{F}, P)$ and $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{P})$. For all $n \in \mathbb{N}$ and $k = 0, \dots, n$ let $\mathcal{A}_{n,k} = \sigma(\bar{S}_{n,i} | i \leq k)$.

From (4) and (7) our error takes the form

$$e_n := e^{-rT} |E[f(S_T)] - \bar{E}[f(\bar{S}_{n,n})]| \quad (19)$$

Since $f(\bar{S}_{n,n}) = c(T, \bar{S}_{n,n})$ we can write

$$\begin{aligned} \hat{E}[f(\bar{S}_{n,n})] &= \hat{E}[c(T, \bar{S}_{n,n})] \text{ and} \\ e^{-rT} \hat{E}[f(S_T)] &= c(0, S_0) = \hat{E}[c(0, S_0)] \end{aligned}$$

Therefore we have

$$e_n = |\hat{E}[e^{-rT} c(T, \bar{S}_{n,n}) - c(0, S_0) | \mathcal{A}_{n,k}]|$$

⁶ At first glance an easy description of the order of convergence seems possible as follows: with the theorems of Berry (1941) and Esséen (1945) the order of convergence of the approximation of a normal by a binomial distribution-function can be calculated. Then, immediately by formulas (3) and (9) we get the order of convergence of e_n . However, from this we get the order $\frac{1}{2}$ for CRR, whereas we observe higher order of convergence in simulations. Other approaches, such as that by Butzer *et al.* (1975) apply a Taylor-expansion. However, this approach is not applicable since the pay-off function of the European call option is not differentiable at $S = K$. In our proof we proceed in a very specific way to circumvent this problem. It relies on the observation that for times $t < T$ $c(t, \cdot) \in C^\infty(\mathbb{R})$. Therefore we apply a trick used by Kloeden and Platen (1992) to a different setting: a discretization of the time-axis and the representation of the error e_n as a sum of relative errors at each time t_i^n .

Discretization of time-axis

$$e_n = \left| \hat{E} \left[\sum_{k=0}^{n-1} e^{-rt_k^n} \{ e^{-r\Delta t} c(t_{k+1}^n, \bar{S}_{n,k+1}) - c(t_k^n, \bar{S}_{n,k}) \} \middle| \mathcal{A}_{n,k} \right] \right|$$

The Black–Scholes price is locally riskless, which means

$$\hat{E}[e^{-r\Delta t} c(t_{k+1}^n, R_{n,k+1} \bar{S}_{n,k}) - c(t_k^n, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] = 0$$

Now it follows

$$\begin{aligned} e_n &= \left| \hat{E} \left[\sum_{k=0}^{n-1} e^{-rt_{k+1}^n} \{ \hat{E}[c(t_{k+1}^n, \bar{S}_{n,k+1}) | \mathcal{A}_{n,k}] - \hat{E}[c(t_{k+1}^n, \bar{S}_{n,k} R_{n,k+1}) | \mathcal{A}_{n,k}] \} \right] \right| \\ &= \left| \hat{E} \left[\sum_{k=0}^{n-1} e^{-rt_{k+1}^n} \{ \hat{E}[c(t_{k+1}^n, \bar{S}_{n,k+1}) - c(t_{k+1}^n, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \right. \right. \\ &\quad \left. \left. - \hat{E}[c(t_{k+1}^n, \bar{S}_{n,k} R_{n,k+1}) - c(t_{k+1}^n, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \} \right] \right| \end{aligned}$$

The last time point $k = n - 1 \Leftrightarrow t_{k+1}^n = T$ must be evaluated separately as $\mathcal{O}(1/n)$. This is an easy, however lengthy and technical, matter.

All other time points $0 \leq k < n - 1$ can be evaluated by a one-dimensional Taylor series expansion of $c(t_k^n, \cdot)$ around $\bar{S}_{n,k}$. If we denote its remainder term by R_3 , the error e_n is bounded by $\mathcal{O}(1/n)$ plus the following term

$$\begin{aligned} &\left| \hat{E} \left[\sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \left\{ \hat{E} \left[\frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k+1} - \bar{S}_{n,k}) + \frac{\partial^2 c}{\partial S^2}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k+1} - \bar{S}_{n,k})^2 \right. \right. \right. \right. \\ &\quad \left. \left. + \frac{\partial^3 c}{\partial S^3}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k+1} - \bar{S}_{n,k})^3 + R_3(t_{k+1}^n, \bar{S}_{n,k+1}, \bar{S}_{n,k}) \middle| \mathcal{A}_{n,k} \right] \right. \\ &\quad \left. - \hat{E} \left[\frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k} R_{n,k+1} - \bar{S}_{n,k}) + \frac{\partial^2 c}{\partial S^2}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k} R_{n,k+1} - \bar{S}_{n,k})^2 \right. \right. \\ &\quad \left. \left. + \frac{\partial^3 c}{\partial S^3}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k} R_{n,k+1} - \bar{S}_{n,k})^3 + R_3(t_{k+1}^n, \bar{S}_{n,k} R_{n,k+1}, \bar{S}_{n,k}) \middle| \mathcal{A}_{n,k} \right] \right\} \right] \right| \end{aligned}$$

One has

$$\begin{aligned} \hat{E} \left[\frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k+1} - \bar{S}_{n,k}) \middle| \mathcal{A}_{n,k} \right] &= \bar{S}_{n,k} \frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k}) \hat{E}[\bar{R}_{n,k+1} - 1] \\ \hat{E} \left[\frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k})(\bar{S}_{n,k} \cdot R_{n,k+1} - \bar{S}_{n,k}) \middle| \mathcal{A}_{n,k} \right] &= \bar{S}_{n,k} \frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k}) \hat{E}[\bar{R}_{n,k+1} - 1] \end{aligned}$$

Analogous results can be obtained for terms with functions $S^2 \frac{\partial^2 c}{\partial S^2}(\cdot, S)$ and $S^3 \frac{\partial^3 c}{\partial S^3}(\cdot, S)$. Therefore we can bound e_n by $\mathcal{O}(1/n)$ plus the following

$$\left| \hat{E} \left[\sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \left\{ \bar{S}_{n,k} \frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k}) \mathbf{m}_n^1 + \bar{S}_{n,k}^2 \frac{\partial^2 c}{\partial S^2}(t_{k+1}^n, \bar{S}_{n,k}) \mathbf{m}_n^2 \right. \right. \right. \\ \left. \left. \left. + \bar{S}_{n,k}^3 \frac{\partial^3 c}{\partial S^3}(t_{k+1}^n, \bar{S}_{n,k}) \mathbf{m}_n^3 + \bar{E}[R_3(t_{k+1}^n, \bar{S}_{n,k+1}, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \right. \right. \right. \\ \left. \left. \left. - \hat{E}[R_3(t_{k+1}^n, \bar{S}_{n,k}, R_{n,k+1}, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \right\} \right] \right|$$

By a detailed (very lengthy and technical) observation of the derivatives one can prove

$$\left| \sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \hat{E} \left[\bar{S}_{n,k} \frac{\partial c}{\partial S}(t_{k+1}^n, \bar{S}_{n,k}) \right] \right| = \mathcal{O}(n) \\ \left| \sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \hat{E} \left[\bar{S}_{n,k}^2 \frac{\partial^2 c}{\partial S^2}(t_{k+1}^n, \bar{S}_{n,k}) \right] \right| = \mathcal{O}(n) \\ \left| \sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \hat{E} \left[\bar{S}_{n,k}^3 \frac{\partial^3 c}{\partial S^3}(t_{k+1}^n, \bar{S}_{n,k}) \right] \right| = \mathcal{O}(n) \\ \left| \hat{E} \left[\sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \hat{E}[R_3(t_{k+1}^n, \bar{S}_{n,k+1}, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \right] \right| = \mathcal{O}(\mathbf{p}_n \cdot n) \\ \left| \hat{E} \left[\sum_{k=0}^{n-2} e^{-rt_{k+1}^n} \hat{E}[R_3(t_{k+1}^n, \bar{S}_{n,k} \cdot R_{n,k+1}, \bar{S}_{n,k}) | \mathcal{A}_{n,k}] \right] \right| = \mathcal{O}\left(\frac{1}{n}\right)$$

These proofs and the evaluation of the last time point – mentioned above – are available upon request from the authors.

If we set $v_n := \exp\{\sigma^2 \frac{T}{n}\}$ we know that $E[R_{n,1}^k] = r_n^k v_n^{k(k-1)/2}$. Applying the binomial formulas $(x+y)^k = \sum_{j=0}^k \binom{k}{j} x^j y^{k-j}$ gives immediately the following equivalent form of Theorem 1, which is useful for applications.

Theorem 2

Conditions as in Theorem 1. Let $\bar{\mathbf{m}}_n^2 := \bar{E}[(\bar{R}_{n,1})^2] - r_n^2 v_n$, $\bar{\mathbf{m}}_n^3 := \bar{E}[(\bar{R}_{n,1})^3] - r_n^3 v_n^3$. Then the conclusions of Theorem 1 hold with $\bar{\mathbf{m}}_n^2, \bar{\mathbf{m}}_n^3$ instead of $\mathbf{m}_n^2, \mathbf{m}_n^3$.

In the case where $\mathbf{m}_n^2, \mathbf{m}_n^3$ have the same order of convergence, exactly the same order of convergence is obtained for $\bar{\mathbf{m}}_n^2, \bar{\mathbf{m}}_n^3$. It turns out that $\mathbf{m}_n^2, \mathbf{m}_n^3$ have the same order of convergence in general. Therefore, we will refer to $\bar{\mathbf{m}}_n^2, \bar{\mathbf{m}}_n^3$ as moments, too. From the context it will always be clear which form should be used. Notably, we proved only that the order of convergence equals

at least one, and possibly a higher order could be obtained, although simulations indicate entirely the opposite. Therefore, Theorems 1 and 2 are a suitable extension of the convergence analysis to the case under consideration here, giving the order of convergence for the price of a European call option. Since the CLT provides the basis of each convergence result, this allows us an analysis of the determinants of the order of convergence. To achieve order of convergence one, the theorem states that the approximating moments of the discrete asset dynamics must converge with order two towards the moments of the continuous asset dynamics, because one degree is lost with summation over time. Furthermore, one needs the same order of convergence in the pseudo-moment⁷.

With the theorem we obtain a characterization of the error boundary which only relies, for all binomial models, on equally well measurable properties. Thus, this is a unique criterion for comparing these models with respect to convergence speed. The theorem reduces the answer to the order of convergence of the moments and pseudo-moments. The results of the theorem can be readily applied to the binomial models under consideration.

Proposition 1

The lattice-approach proposed by Cox, Ross and Rubenstein (1979) converges with order 1.

Proposition 2

The lattice-approach proposed by Jarrow and Rudd (1983) converges with order 1.

Proposition 3

The lattice-approach proposed by Tian (1993) converges with order 1.

The proofs of these propositions proceed by application of Theorem 2, through an examination of (pseudo-) moments in a straightforward way. Therefore, they are postponed to Appendix B.

In Figs 5 to 7 we depict a simulation using the CRR, JR and Tian lattice approaches. The left-hand figures show the error with the typical wavy patterns. The pictures on the right-hand side depict the (pseudo-) moments. Please note that the lattice approach by Tian is constructed such that the first three moments are equal to 0. Remarkably for all models, the convergence behaviour of the (pseudo-) moments is very smooth and even comparing it with the sequence $(1/n^2)_n$ we see no difference between them. Thus, the order of convergence of (pseudo-) moments can be deduced easily through simulations. According to Theorem 2, we therefore have order one for convergence of prices. Comparing the convergence behaviour on the left and right-hand sides of each figure, we note that determining the order of convergence for prices through (pseudo-) moments and our theorems is much easier than through the underlying error picture.

Therefore, both the methods by proof (see Appendix B) and by simulation for determining the order of convergence from that of (pseudo-) moments, work very well.

Please note that although Theorems 1 and 2 do not allow any statement about the constant κ the simulations in the figures for the different strikes seem to suggest that, at least for the lattices and strikes under consideration here, they are about the same.

⁷ This is not only a technical matter but explains why the model proposed by Tian does not perform better.

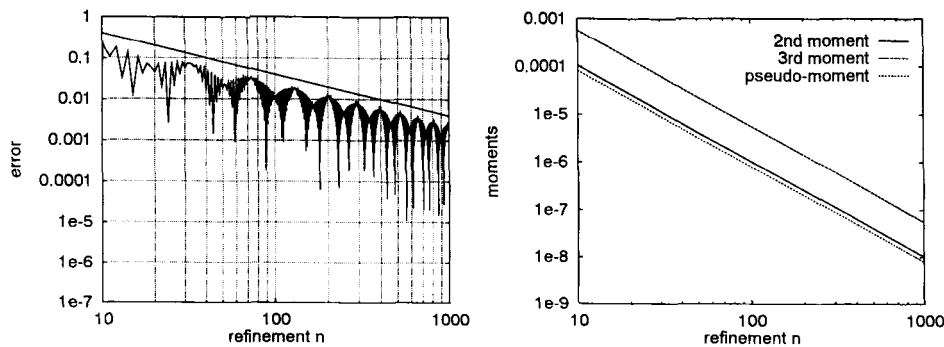


Fig. 5. Illustration of the result of Proposition 1: order of convergence one because second and third moment and pseudo-moment have order of convergence two: example with CRR-model and the following selection of parameters: $S = 100$, $K = 90$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 1000$.

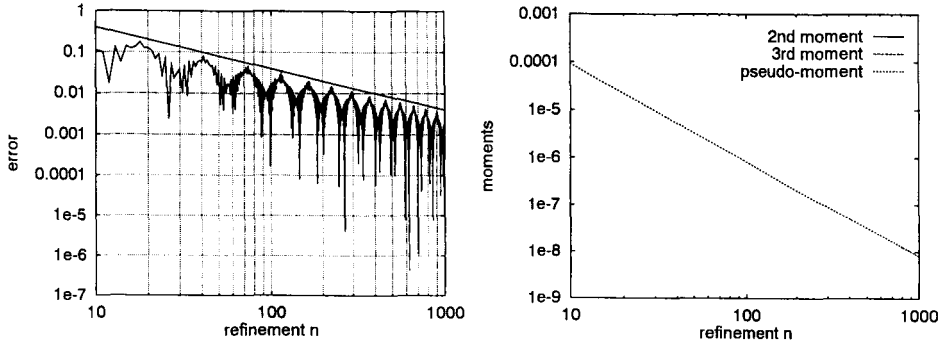


Fig. 6. Illustration of the result of Proposition 2: order of convergence one because pseudo-moment has order of convergence two: example with Tian-model and the following selection of parameters: $S = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 1000$.

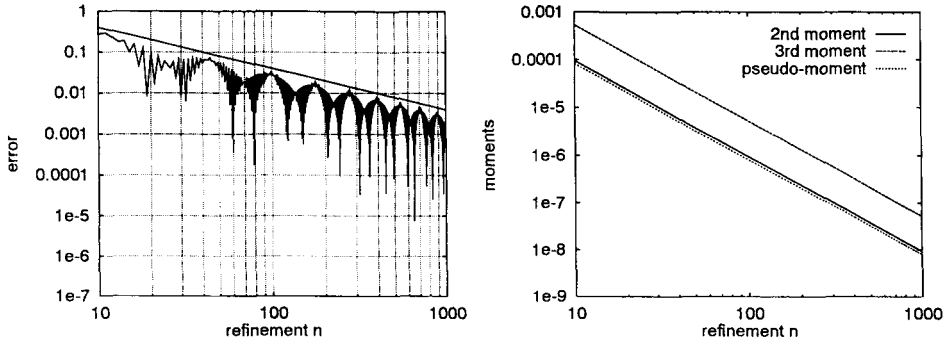


Fig. 7. Illustration of the result of Proposition 3: order of convergence one because second and third moment and pseudo-moment have order of convergence two: example with JR-model and the following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 1000$.

It can be shown that explicit finite difference methods, which approximate the deterministic Black–Scholes price function (3) through a discretization of the Black–Scholes partial differential equation (2) converge with order one, too. Bearing in mind the connection between this method and lattice approaches mentioned at the end of the previous section our result is consistent with this finding; we see it thus confirmed.

4. Construction of binomial models with improved convergence properties

So far, we have examined the problem of convergence behaviour and speed in lattice approaches from the literature. We derived a general theorem to measure the order of convergence. Applying the theorem to the considered binomial models we see no significant difference in terms of convergence to the Black–Scholes solution. In this part of the paper we present a method to construct improved approximating trees for the calculation of option prices. Here, our aim is twofold: on the one hand, we desire to obtain binomial trees without irregularities, and on the other we want faster convergence speed. Unfortunately, the theorem indicates that improved order of convergence as resulting from the properties of the binomial process would require higher order of convergence for the moments and pseudo-moment. In principle, this cannot be achieved with a two-point random variable. Now, the only possible recipe consists of utilizing structural properties of the considered option contract.

We call the resulting method an *extended lattice approach* to emphasize the difference to the standard lattice approach.

CRR showed convergence of their model to the Black–Scholes formula by examining the two binomial terms in (9) separately. Equivalently, one can view the valuation of a European call option as two approximations of type $\Phi[a; n, p] \rightarrow \mathcal{N}(z)$. We take this approximation task as our starting point for improvements.

Generally speaking, despite the simplicity, the calculation of binomial probabilities is cumbersome because it might involve factorials of large integers or the summation of a large number of individual terms. Therefore, normal approximations to the binomial distribution were derived in the mathematical literature. In particular, the method by Camp (1951) and Paulson (1942) and the approximations of Peizer and Pratt (1968) reveal a remarkable precision⁸. With all these approaches, basically a binomially calculated true probability P is approximated with the standard normal function $\mathcal{N}(z)$, where the input is determined by some adjustment function $z = h(a; n, p)$ where, in our setting, a is the number of up-movements of the asset price to exceed the strike price in an n -step binomial tree with martingale measure equal to p . In the simplest case we have the de Moivre–Laplace theorem. Here $P = 1 - \Phi[a; n, p]$ is approximated by $P \approx \mathcal{N}(h(a; n, p)) = (a - np)/(np(1 - p))^{1/2}$, where $\Phi[a; n, p]$ is the complementary binomial distribution function of (9).

However, our option pricing problem represents the opposite direction: computation of binomial option prices involves the fact that normal components are approximated by binomial components. Thus, for a given binomial tree refinement inverting the adjustment function, $h(a; n, p)$ above specifies the distribution parameter $h^{-1}(z) = p$ to approximate $P = \mathcal{N}(z)$ with

⁸ The reader will find some remarks as to the derivation of these approximations and the citation of literature in Appendix A.

$P \approx 1 - \Phi[a; n, p]$. Peizer and Pratt derived the inversion formula to the Camp–Paulson method and specified the inversion formula of their method in the case with an identical number of successes and failures⁹.

(A) *Camp–Paulson inversion*

$$h^{-1}(z) = \left(\frac{y}{x}\right)^2 \left(\frac{[9x-1][9y-1] + 3z[x(9y-1)^2 + y(9x-1)^2 - 9xyz^2]^{\frac{1}{2}}}{[9y-1]^2 - 9yz^2} \right)^{\frac{1}{3}}$$

with $x = n - a$, $y = a + 1$, z as the input of the standard normal function.

(B) *Peizer–Pratt method 1 inversion* [case: $a + \frac{1}{2} = n - (a + \frac{1}{2})$, $n = 2a + 1$]

$$h^{-1}(z) = 0.5 \mp \left[0.25 - 0.25 \exp \left\{ - \left(\frac{z}{n + \frac{1}{3}} \right)^2 \left(n + \frac{1}{6} \right) \right\} \right]^{\frac{1}{2}}$$

(C) *Peizer–Pratt method 2 inversion* [case : $a + \frac{1}{2} = n - (a + \frac{1}{2})$, $n = 2a + 1$]

$$h^{-1}(z) = 0.5 \mp \left[0.25 - 0.25 \exp \left\{ - \left(\frac{z}{n + \frac{1}{3} + \frac{0.1}{(n+1)}} \right)^2 \left(n + \frac{1}{6} \right) \right\} \right]^{\frac{1}{2}}$$

So far, we have discussed approximating normal through binomial distributed terms. Now, we demonstrate how these findings can be used to construct CRR-like binomial models. We set up an equation system to determine uniquely the tree parameters which ensure convergence. First, as there are two components in the pricing formulas (3) and (9), with d_1 and d_2 as inputs from equation (3) in $h^{-1}(z)$, we obtain p and p' as distribution parameters of the two binomial components in the binomial option pricing formula using approximation rule A, B or C.

Then, we derive tree parameters u_n and d_n by a simple trick. The no-arbitrage condition implies that $p_n = (r_n - d_n)/(u_n - d_n)$ holds. Furthermore, p'_n is defined by $p'_n = u_n/r_n \cdot p_n$. Taking these two relations as a system of equations that can be solved uniquely with respect to u_n and d_n , we obtain a new binomial model. In summary¹⁰

$$p'_n = h^{-1}(d_1)$$

⁹ Otherwise the inversion could be achieved numerically. Since this inversion is generally valid to any parameter selection, it could be tabulated or approximated polynomially for fixed n in a manner similar to the way values of the standard normal distribution function are tabulated or approximated.

¹⁰ In the paper here, we consider only three inverted normal approximations. The method to construct a binomial lattice is general with respect to the chosen normal approximation. In particular one could invert the simple continuity correction

$$P \approx \mathcal{N} \left(h(a; n, p) = \frac{a - 0.5 - np}{(np(1-p))^{1/2}} \right)$$

and apply it as an approximation rule.

$$p_n = h^{-1}(d_2)$$
$$u_n = r_n \frac{p'_n}{p_n}$$
$$d_n = \frac{r_n - p_n u_n}{1 - p_n}$$

Note that having parameter z , for each pair of (a, n) we get a unique distribution parameter p . Consequently, from the perspective of the continuous model the parameter $a(n)$ can be chosen freely. In the system of equations above for the determination of p_n and p'_n , $a(n)$ and $a'(n)$ can be chosen freely but must be equal. We choose $a(n)$ such that the strike price K is located in the centre of the binomial tree. Regarding the fact that the usual interest focuses almost entirely on at-the-money and near-the-money options, this selection does not distort the tree significantly.

In example calculations, the resulting binomial tree parameters diverge only very little from those of previous models but, astonishingly, the convergence properties with the computation of

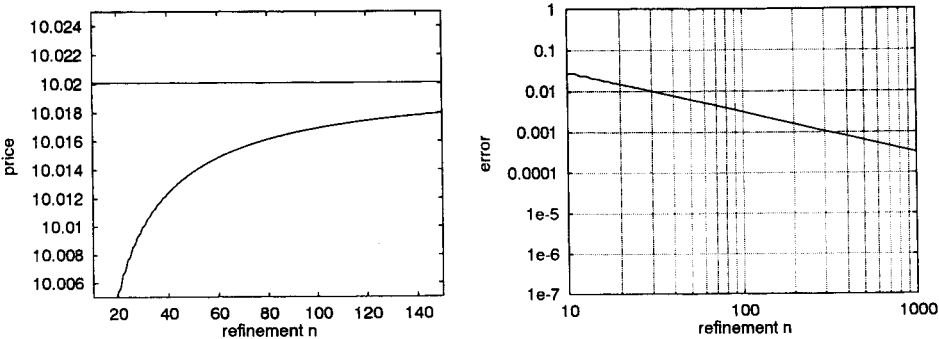


Fig. 8. Illustration of the results with the improved binomial construction theme using the inverted normal approximation of Camp–Paulson; following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 150$ ($n = 10, \dots, 1000$).

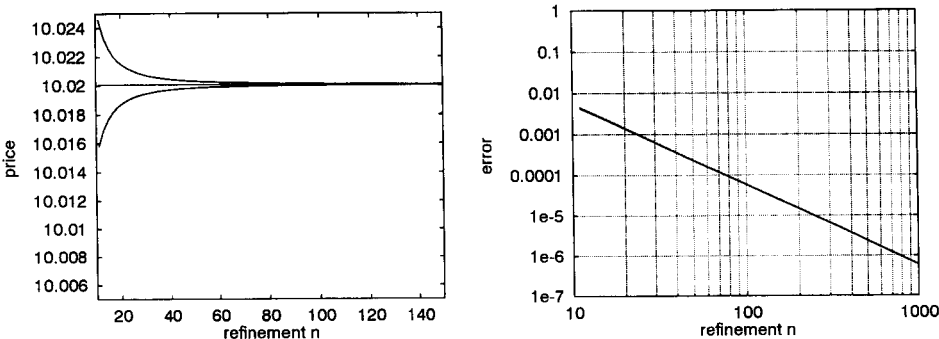


Fig. 9. Illustration of the results with the improved binomial construction theme using both inverted normal approximations of Peizer–Pratt; following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 150$ ($n = 10, \dots, 1000$).

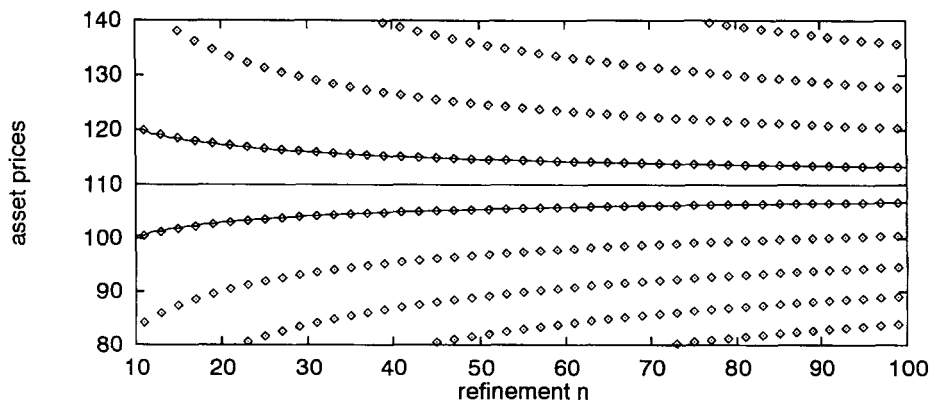


Fig. 10. Terminal asset price grids with iteration of the tree refinement and location of the strike price: example with PP2-model and the following selection of parameters: $S = 100$, $K = 110$, $T = 1$, $r = 0.05$, $\sigma = 0.3$, $n = 10, \dots, 100$.

option prices change tremendously¹¹. This model construction profits from the attributes of the chosen normal approximation. The figures below demonstrate the strength of the method in approximating option prices in comparison to previously existing models. Figure 8 shows an example of the results for the binomial model using the Camp–Paulson method (method (A)). Notably, the y -axis of the left-hand figure has a range of two cents. The approximation error declines monotonically with order of convergence one and the constant κ is approximately equal to one instead of four (compare with Figs 5, 6, and 7). Figure 9 depicts an example of the results with the Peizer–Pratt methods (methods (B) and (C)). The two methods do not show significant differences in the application. Here, we take odd refinements only due to the restricted domain of the explicit inversion formula. This does not diminish the results because there is no meaning to any specific refinement except serving for approximation purposes. Again the approximation error declines monotonically but now with order of convergence, ρ , equal to two and constant, κ , (see (14)) approximately equal to one.

Remember our reflections about the strike location problem in Section 3. Here, a strike location at points across the streamlines with changing distances to surrounding tree nodes cannot result. The location of the strike within the tree grid is controlled (see Fig. 10). Notably, the accuracy of the reference price calculated with (3) relies on the accuracy of the applied method for calculating the normal distribution function. We experienced distorted results when using simple approximations for the standard normal distribution function with too few digits of precision.

Indeed, the use of the superior inverted normal approximation improves the calculation of option prices. The performance here allows for a considerable reduction of the tree refinement to save computation time. Looking at the moments and pseudo-moments, the error declines with order of convergence one only. This discloses that the improvement is achieved by utilization of

¹¹ Nevertheless, within this class of models the particular theoretical building blocks for which the CRR-model became famous are entirely preserved by construction. We have completeness and a unique replicating strategy can be derived.

structural properties of the chosen contract type. Consequently, we are not able to give a strict proof of the greater order of convergence in line with our theorem.

We obtained our procedure, analysing the valuation of European call- and put options. Now we examine the valuation of the American put option. Amin and Khanna (1994) prove the convergence of the American option prices in the case where the limiting process of the discrete-time model is a diffusion process satisfying convergence with the first two moments and one higher moment. It can be shown that their results readily apply to our model.

Consider a particular lattice approach. For each refinement n , we have a discrete time grid $T^n = \{0 = t_0^n, \dots, t_n^n = T\}$ with $t_{i+1}^n - t_i^n = \frac{T}{n}$ ($i = 0, \dots, n-1$) under consideration. Sequentially comparing prices from backward induction with local inner values, the continuous early exercise boundary B_t^* is then approximated by critical asset prices $(B_{t_i^n}^*)_{i=1, \dots, n}$ where exercise occurs. Now take a fixed refinement m with dates T^m . This needs to be distinguished from T^n , the time grid of a particular binomial lattice, because T^m may be some subset of T^n . Taking $B_{t_i^m}^*$, the American put option can be approximated with a portfolio consisting of the corresponding European put option and a set of binary options¹²

$$c_n^{\text{put}}(0 = t_0^n, S_0) = Kr_n^{-n} \Phi[a; n, p_n] - S_0 \Phi[a; n, p'_n] + rK \sum_{i=0}^m r_m^{-i} \Phi[b_i^{n,m}; n, p_n] \quad (20)$$

where a is defined as before, $b_i^{n,m} = \lfloor (\ln(B_{t_i^m}^*/S_0) - j_i \ln d_n) / (\ln u_n - \ln d_n) \rfloor$ and $j_i = \lfloor in/m \rfloor$

This portfolio represents a binomial translation of the decomposition of Carr *et al.* (1992).

Now, we keep this portfolio with m binary options for a moment and evaluate the speed of convergence with an iteration of n , only. Obviously, our improved scheme holds for the European put component. Now, suppose we had approximated the true early exercise boundary with $(B_{t_i^m}^*)_{i=1, \dots, n}$ at the time points from T^m perfectly. Remember the discussion earlier about the irregularities resulting from the strike separating the probability mass. Even with $(B_{t_i^m}^*)_{i=1, \dots, n}$ we have the same problem again, but for each binary option, i.e. for each time point from T^n with respective strike $B_{t_i^m}^*$. Without specific regard to the strike location in the constructed trees we obtained order of convergence one. Consistently, we state here without proof that, for the kept portfolio of binary options, the order of convergence to

$$Ke^{-r(T-t)} \mathcal{N}(d_2) - S \mathcal{N}(d_1) + rK \sum_{i=0}^m e^{-r(T-t_i^m)} \mathcal{N}(d_{3i})$$

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma(T-t)^{1/2}}$$

$$d_{3i} = \frac{\ln(S/B_{t_i^m}^*) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma(T-t)^{1/2}}, \quad i = 0, \dots, m$$

is equal to one, because with any tree refinement, there is a strike separation problem included along the exercise boundary.

¹² Of course, with the pricing of the American put, the number of binary options grows with the tree refinement, but we isolate the specific approximation task of this portfolio. Actually, keeping one such portfolio and iterating the tree refinement introduces new problems because the time grid no longer resembles the times to maturity in (20). We assume this problem away here (e.g. assume iteration where $w\Delta t_{n+1} = \Delta t_n, w \in \mathbb{N}$ holds).

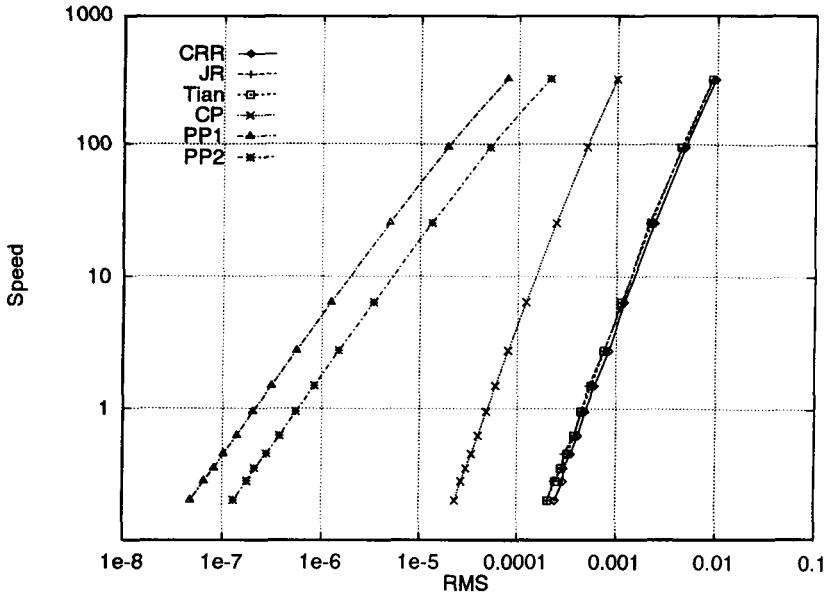


Fig. 11. Testing efficiency of binomial models for European call options: relative mean squared error in calculations with a large parameter sample, sample size $m = 2500$, versus computation speed as expressed by the number of option prices calculated per second, calculations at marked points with tree refinements: $n_i = \{25, 50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$; x -axis and y -axis with log-scale.

Hence, with transition to the approximation valuation of American options, involving $B_{i,n}^*$ instead of $B_{i,n}^*$ and $m \rightarrow \infty$, thus having a portfolio with a growing number of binary options again, we do not expect to obtain an order of convergence better than one. On the other hand, our numerical results presented in the next section indicate that the order of convergence is kept equal to one.

Summarizing, we expect the error with the approximation of American put prices to converge with order one¹³. Nevertheless, a considerable improvement potentially arises, because for the European put component included in the decomposition portfolio the error is considerably reduced. Starting with an approximation from a standard CRR-like model, a net improvement results whenever the error produced along the portfolio of binary options does not increase with the transition to our theme by more than the amount of the European put component improvement. In the table of example calculations (see the next section) only the deep-in-the-money case has net-deterioration, whereas the four other examples exhibit considerable net improvement.

5. Numerical results

Now we compare the three binomial methods considered in Section 2 and 3 with those newly developed in Section 4. Table 2 contains example computations for European call and put

¹³ One could extend our procedure by structural adjustments along $B_{i,n}^*$ in order to obtain faster convergence, where $B_{i,n}^*$ could be determined by quadratic interpolation between exercise and non-exercise nodes.

Table 2. A table with numerical results, a small window of the whole approximation theme with accidental degrees of accuracy, but useful for those readers intending the implementation of methods; parameters $S = 100$, $r = 0.07$, $\sigma = 0.3$, $T = 0.5$ years, $n = 25$ for all.

Strike	CRR	JR	Tian	CP	PPI	PP2	True value
European call options							
80	23.74082	23.76300	23.70657	23.76050	23.75822	23.75875	23.75799
90	16.13376	16.08486	16.12494	16.09619	16.09941	16.10037	16.09963
100	10.21317	10.20142	10.20418	10.12545	10.13316	10.13440	10.13377
110	6.01218	6.02481	6.01304	5.94162	5.94889	5.95015	5.94946
120	3.31890	3.33429	3.33318	3.27993	3.28258	3.28366	3.28280
European put options							
80	0.98926	1.01143	0.95500	1.00893	1.00665	1.00719	1.00642
90	3.03825	2.98934	3.02943	3.00068	3.00390	3.00486	3.00412
100	6.77371	6.76196	6.76472	6.68599	6.69370	6.69494	6.69431
110	12.22878	12.24141	12.22963	12.15821	12.16548	12.16675	12.16606
120	19.19155	19.20694	19.20583	19.15258	19.15523	19.15631	19.15545

options. Even with the very low tree refinement of $n = 25$, the outstanding performance of models using normal approximations can be recognized. Five digits must be displayed to catch the degree of accuracy. Each considered simulation result may depend significantly on an accidentally chosen parameter set. Thus, we looked for a procedure to test simultaneously across a whole set of parameters. We conduct an analysis following an approach of Broadie and Detemple (1996), who tested several methods for the pricing of American options. There, within one analysis, several methods using a large sample of randomly selected parameters are compared simultaneously over refinements with a measurement of computation speed and approximation error. Computation speed is expressed by the number of option prices calculated per second. Since we confine ourselves to tree models with identical structures, except for the tree parameters, we can use the speed results of Broadie and Detemple for CRR, for all the models considered here. The approximation error is measured by the relative root-mean-squared (RMS) error. RMS-error is defined by

$$\text{RMS} = \left(\frac{1}{m} \sum_{i=1}^m e_i^2 \right)^{1/2}$$

where $e_i = (\hat{c}_i - c_i)/c_i$ is the relative error, c_i is the true option value. $\hat{c}_i$ is the estimated option value. To make relative error meaningful, that is to avoid senseless distortions because of very small option prices, the summation is taken over options in the dataset satisfying $c_i \geq 0.50$.

We choose the following distribution of parameters. Volatility is distributed uniformly between 0.1 and 0.6. Time to maturity is, with probability 0.75, uniform between 0.1 and 1.0 years and, with probability 0.25, uniform between 1.0 and 5.0 years. We fix the strike price at $K = 100$ and take the initial asset price $S \equiv S_0$ to be uniform between 70 and 130. Relative errors do not change if S and K are scaled by the same factor, i.e. only the ratio S/K is of interest. The riskless rate r is, with probability 0.8, uniform between 0.0 and 0.10 and, with probability 0.2, equal to 0.0. Each parameter is selected independently of the others. This selection of parameters matches the choice of Broadie and Detemple (1996) except for dividends which we do not consider here.

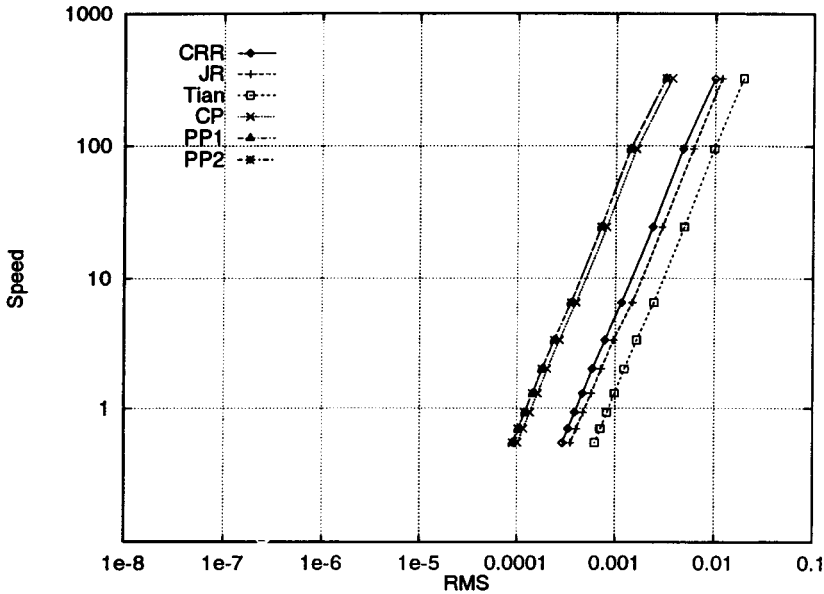


Fig. 12. Testing efficiency of binomial models for American put options: relative mean squared error in calculations with a large parameter sample, sample size $m = 2500$, versus computation speed as expressed by the number of option prices calculated per second, calculations at marked points with tree refinements: $n_i = \{25, 50, 100, 200, 300, 400, 500, 600, 700, 800\}$; x-axis and y-axis with log-scale.

Figure 11 reports the results for European call options to which the analysis was devoted so far. Of course, similar results could be presented for European put options. The newly developed methods outperform all tested approximations in terms of accuracy. According to the figure above, a 25-step-tree computation with our procedure is on average, as accurate as a 1000-step-tree computation with the CRR model. Notice, this means that our procedure is about 1400 times faster! We reproduced the finding of Broadie and Detemple (1996) that the speed-accuracy line for the CRR-model is linear in appearance. These findings carry over to the JR-model, Tian-model, and our approaches. The smooth line in the figure results from the averaging over the results of the whole sample. Taking only a single parameter constellation yields a picture where the convergence patterns described earlier emerge again, whereas the lines for the new methods remain stable. Although the tree adjustment primarily served for the improved approximation of European standard options, we see that valuable improvements for the pricing of American put options also result. True American put option values were derived using the CRR-method using 15000 tree-steps (see Table 3). Figure 12 reports speed-accuracy results for the calculation of American put options. Similar results are obtained for the previous models¹⁴. The new models exhibit

¹⁴ Interestingly, the CRR dominates JR and Tian. With CRR we have the property $ud = 1$. From simulations one can see that the early exercise boundary approaches a constant line with movement towards t_0 in the tree. Consequently, we expect less distorting crossing of node layers and the early exercise boundary.

Table 3. A table with numerical results, a small window of the whole approximation theme with accidental degrees of accuracy, but useful for those readers intending the implementation of methods; parameters $S = 100$, $r = 0.07$, $\sigma = 0.3$, $T = 0.5$ years, $n = 25$ for all.

Strike	CRR	JR	Tian	CP	PP1	PP2	True value
80	1.01842	1.03864	0.98396	1.04231	1.04264	1.04317	1.037
90	3.16580	3.12447	3.14640	3.11786	3.12832	3.12928	3.123
100	7.10823	7.10415	7.08701	7.00982	7.02858	7.02981	7.035
110	13.00108	13.01511	12.98978	12.90304	12.93136	12.93253	12.955
120	20.73344	20.74479	20.73566	20.65254	20.67576	20.67649	20.717

order of convergence one but with much smaller initial error. According to the figure above, a 100-step-tree computation with our procedure is as accurate, on average, as a 300-step-tree computation with the CRR-model. Notice that this means that our procedure is 9 to 10 times faster! The observed improvement confirms that the approximation error chiefly arises from deficiencies in connection with the terminal pay-off distribution.

6. Conclusions

In this paper we compare different lattice approaches with respect to their convergence behaviour and speed. Concerning the convergence behaviour, we can explain the sources of the occurring wavy patterns. The convergence speed we measure by the order of convergence. We prove a general theorem which allows an easy derivation of the order of convergence with particular models under consideration. For the three different approaches from the literature considered here, the result of this analysis turns out to be fairly disappointing, since all models converge with order one and need to be seen therefore as equivalent. In the next step, using structural properties we construct a lattice approach converging with the higher order two and a lower constant.

To some extent the findings of this analysis carry over to American put options. The existing lattice approaches seem to hold order one. Our method seems to have the same order but a much lower constant. The reason for this and potential ways to resolve this problem are explained. Yet here remains work for future research.

References

- Abramowitz, M. and Stegun, I. (Eds) (1968) *Handbook of Mathematical Functions*, 5th Edition, Dover Printing.
- Amin, K. (1991) On the computation of continuous time option prices using discrete approximations, *J. Financial and Quantitative Analysis*, **26**, 477–96.
- Amin, K. and Khanna, A. (1994) Convergence of American option values from discrete- to continuous-time financial models, *Mathematical Finance*, **4**, 289–304.
- Berry, A.C. (1941) The accuracy of the Gaussian approximation to the sum of independent variables, *Trans. Amer. Math. Soc.*, **49**, 122–36.
- Boyle, P.P. (1988) Option valuation using a three-jump process, *Int. Options J.*, **3**, 7–12.

- Boyle, P.P., Evnine, J. and Gibbs S. (1989) Numerical evaluation of multivariate contingent claims, *The Review of Financial Studies*, **2**, 241–50.
- Broadie, M. and Detemple, J. (1996) American option evaluation: new bounds, approximations, and a comparison of existing methods, *Review of Financial Studies*, **9**, to appear.
- Butzer, P.L., Hahn, L. and Westphal, U. (1975) On the rate of approximation in the central limit theorem, *J. Appl. Theory*, **13**, 327–40.
- Camp, B.H. (1951) Approximation to the point binomial, *Annals of Mathematical Statistics*, **22**, 130–31.
- Cheuk, T.H.F. and Vorst, T.C.F. (1994) Binomial models for some path-dependent options, Working Paper Tinbergen Institute, Erasmus University Rotterdam, Report 9422/A.
- Cox, J., Ross, S.A. and Rubinstein M. (1979) Option pricing: a simplified approach, *J. Financial Economics*, **7**, 229–63.
- Cox, J. and Rubinstein, M. (1985) *Options Markets*, Prentice-Hall, New Jersey.
- Esséen, C.G. (1945) Fourier analysis of distribution functions, *Acta Math.*, **77**, 1–125.
- Fieller, E.C. (1932) The distribution of the index in a normal bivariate population, *Biometrika*, **24**, 428–40.
- Geary, R.C. (1930) The frequency distribution of the quotient of two normal variates, *J. Roy. Stat. Soc.*, **93**, 442–6.
- Gebhardt, F. (1960) Some numerical comparisons of several approximations to the binomial distribution, *J. Am. Stat. Assoc.*, **64**, 1638–46.
- Geske, R. and Shastri, K. (1985) Valuation by approximation: a comparison of alternative option valuation techniques, *J. Financial and Quantitative Analysis*, **20**, 45–71.
- Harrison, J. and Pliska S. (1981) Martingales and stochastic integrals in the theory of continuous trading, *Stochastic Processes and their Applications*, **11**, 215–60.
- He, H. (1990) Convergence from discrete- to continuous-time contingent claims prices, *Rev. Financial Studies* **3**, 523–46.
- Hull, J. and White, A. (1993) Efficient procedures for valuing path-dependent options, Working Paper, Faculty of Management University of Toronto.
- Kendall, M. and Stuart, A. (1977) *The Advanced Theory of Statistics*, Vol. 1, Charles Griffin & Company, London.
- Kloeden, P.E. and Platen, E. (1992) *Numerical Solution of Stochastic Differential Equations*, Springer Verlag.
- Li, A. (1992) Binomial approximation: computational simplicity and convergence, Working Paper #9201, Federal Reserve Bank of Cleveland.
- Madan, D.B., Milne, F. and Shefrin, H. (1989) The multinomial option pricing model and its Brownian and Poisson limits, *Rev. of Financial Studies*, **2**, 251–65.
- Nelson, D.B. and Ramaswamy, K. (1990) Simple binomial processes as diffusion approximations in financial models, *Rev. of Financial Studies*, **3**, 393–430.
- Omberg, E. (1988) Efficient discrete time jump process models in option pricing, *J. Financial and Quantitative Analysis*, **23**, 161–74.
- Paulson, E. (1942) An approximate normalization of the analysis of variance distribution, *Ann. Mathematical Statistics*, **13**, 233–5.
- Peizer, D.B. and Pratt, J.W. (1968) A normal approximation for binomial, F, beta, and other common related tail probabilities, I, *J. Am. Statist. Assoc.*, **63**, 1416–56.
- Pratt, J.W. (1968) A normal approximation for binomial, F, beta, and other common, related tail probabilities, II, *J. Am. Statist. Assoc.* **63**, 1457–83.
- Raff, M.S. (1955) The comparative accuracy of several approximations to the cumulative binomial distribution, Unpublished Master's thesis. Washington: The American University.
- Raff, M.S. (1956) On approximating the point binomial, *Am. Statist. Assoc. J.* June.
- Rendleman, R. and Bartter, B. (1979) Two-state option pricing, *J. Finance*, **34**, 1093–1110.
- Rubinstein, M. (1990) Exotic options, Working Paper University of California at Berkeley.

- Tian, Y. (1993) A modified lattice approach to option pricing, *J. Futures Markets*, **13**, 563–577.
- Wilmott, P., Dewynne, J. and Howison, S. (1993) *Option Pricing: Mathematical Models and Computation*, Oxford Financial Press.
- Wilson, E.B. and Hilferty, M.M. (1931) The distribution of chi-square, *Proc. National Academy of Sciences*, **17**, 684–88.

Appendix A: Normal approximations

*Camp–Paulson approximation*¹⁵

This approximation is based on the equivalence of a cumulative binomial probability to an incomplete beta-function ratio (Kendall and Stuart, (1977), p. 131) and thereby to a probability integral of the variance ratio, F (Kendall and Stuart, (1977), p. 407). Using an approximation to the integral of F developed by Paulson (1942) (Kendall and Stuart, (1977), p. 410), who in turn used Wilson and Hilferty's (1931) approximation for the distribution of chi-square (Kendall and Stuart, (1977), p. 399) and the result obtained by Fieller (1932) and Geary (1930) concerning the ratio of two normally distributed variates (Kendall and Stuart (1977), p. 288), Camp (1951) developed an explicit expression which may be written as follows

$$B(a, n; p) = \mathcal{N}\left(\frac{-x_1}{3\sqrt{x_2}}\right)$$

$$x_1 = \left[(n-a)\frac{p}{a+1}(1-p)\right]^{\frac{1}{3}}\left[9 - \frac{1}{n-a}\right] + \frac{1}{a+1}$$

$$x_2 = \left[n-a\frac{p}{a+1}(1-p)\right]^{\frac{2}{3}}\frac{1}{n-a} + \frac{1}{a+1}$$

see also Gebhardt (1969) and Peizer and Pratt (1968). Peizer and Pratt (1968) derived the inversion formula presented in Section 4.

*Peizer–Pratt approximations*¹⁶

Let z be the true but functionally unknown input of the standard normal function to approximate a value of the cumulative binomial distribution function. Starting with approximation $z^* = [(a + \frac{1}{2}) - np]/\sqrt{npq}$, where $a + \frac{1}{2}$ denotes the number of successes in n Bernoulli trials with

¹⁵ This description is obtained mainly from Raff (1956) and, especially, references to Kendall and Stuart (1977) were supplemented.

¹⁶ See Peizer and Pratt (1968) and Pratt (1968) for these approximations.

continuity correction, correct for misplacement of the median by

$$\frac{z^* = \left[a + \frac{1}{2} - np + \frac{q-p}{6} \right]}{\sqrt{npq}}$$

and further investigations suggest replacing n by $n + \frac{1}{6}$ in the denominator. A thorough investigation of approximation patterns with z/z^* reveals functionally expressible simple patterns, which eventually lead to the following adjustment

$$z_1 = \frac{\left[(a + \frac{1}{2}) - np + \frac{q-p}{6} \right]}{((n + \frac{1}{6})pq)^{1/2}} \underbrace{\left[1 + qg\left(\frac{a + \frac{1}{2}}{np}\right) + pg\left(\frac{n - (a + \frac{1}{2})}{nq}\right) \right]}_G^{\frac{1}{2}}$$

where $g(x) = (1-x)^{-2}(1-x^2+2x\ln x)$

Further modification to the first part delivers a second approximation

$$z_2 = \frac{\left\{ \left[(a + \frac{1}{2}) - np + \frac{q-p}{6} \right] + 0.02 \left(\frac{q}{a+1} - \frac{p}{n-a} \frac{q-0.5}{n+1} \right) \right\}}{((n + \frac{1}{6})pq)^{1/2}} G$$

In the case where $a + \frac{1}{2} = n - (a + \frac{1}{2})$ these formulas reduce to

$$z_1 = \pm \left(n + \frac{1}{3} \right) \left\{ \frac{-\ln(4pq)}{n + \frac{1}{6}} \right\}^{\frac{1}{2}}$$

$$z_2 = \pm \left(n + \frac{1}{3} \right) \left\{ \frac{-\ln(4pq)}{n + \frac{1}{6}} \right\}^{\frac{1}{2}}$$

where the sign is to be chosen to agree with the sign of $q - 0.5$.

Only then can inversion formulas as presented in Section 4 be derived.

Appendix B: Proofs of Propositions 1, 2, and 3

We remember $r_n := \exp\{r\Delta t_n\}$, $v_n := \exp\{\sigma^2\Delta t_n\}$ and for all $k \in \mathbb{N}$: $E[R_{n,1}^k] = r_n^k v_n^{k(k-1)/2}$.

Proof of Proposition 1

From Theorem 2 we know that we only have to verify the following three terms:

$$(1) \quad \bar{\mathbf{m}}_n^2 = \mathcal{O}\left(\frac{1}{n^2}\right)$$

Since $p_n u_n + q_n d_n = r_n$ one gets

$$\begin{aligned} p_n u_n^2 + q_n (u_n d_n) &= u_n r_n \text{ and } p_n (u_n d_n) + q_n d_n^2 = r_n d_n \\ \Rightarrow p_n u_n^2 + q_n d_n^2 &= r_n (u_n + d_n) - (u_n d_n)(p_n + q_n) = r_n (u_n + d_n) - 1 \\ \Rightarrow \bar{\mathbf{m}}_n^2 &= |p_n u_n^2 + q_n d_n^2 - r_n^2 v_n| = r_n |u_n + d_n - r_n^{-1} - r_n v_n| \\ &= r_n \mathcal{O}\left(\frac{1}{n^2}\right) \\ &= \mathcal{O}\left(\frac{1}{n^2}\right) \end{aligned}$$

from a series expansion of the exponential function.

$$(2) \quad \bar{\mathbf{m}}_n^3 = \mathcal{O}\left(\frac{1}{n^2}\right)$$

is proven the same way.

$$(3) \quad \mathbf{p}_n = \mathcal{O}\left(\frac{1}{n^2}\right)$$

From a series expansion of the exponential function one immediately gets

$$u_n - 1 = \mathcal{O}\left(\frac{1}{\sqrt{n}}\right), \quad d_n - 1 = \mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$$

Since $\ln u_n = -\ln d_n = \sigma\sqrt{\Delta t_n}$ one gets $\mathbf{p}_n = \mathcal{O}\left(\frac{1}{n^2}\right)$

Proof of Proposition 2

From Theorem 2 we know that we only have to verify the following three terms:

$$(1) \quad \bar{\mathbf{m}}_n^2 = \mathcal{O}\left(\frac{1}{n^2}\right)$$

As in the previous theorem one gets

$$\begin{aligned} p_n u_n^2 + q_n d_n^2 &= r_n (u_n + d_n) - (u_n d_n) = r_n (u_n + d_n) - e^{2\mu' \Delta t_n} \\ \Rightarrow \bar{\mathbf{m}}_n^2 &= |p_n u_n^2 + q_n d_n^2 - r_n^2 v_n| \\ &= r_n e^{\mu' \Delta t_n} |u_n e^{-\mu' \Delta t_n} + d_n e^{-\mu' \Delta t_n} - e^{\mu' \Delta t_n} e^{-r \Delta t_n} e^{-(2r + \sigma^2) \Delta t_n}| \\ &= r_n e^{\mu' \Delta t_n} \cdot \mathcal{O}\left(\frac{1}{n^2}\right) \\ &= \mathcal{O}\left(\frac{1}{n^2}\right) \end{aligned}$$

by a series expansion of the exponential function.

$$(2) \quad \bar{\mathbf{m}}_n^3 = \mathcal{O}\left(\frac{1}{n^2}\right)$$

is proven the same way.

$$(3) \quad \mathbf{p}_n = \mathcal{O}\left(\frac{1}{n^2}\right)$$

From a series expansion of the exponential function one immediately gets

$$u_n - 1 = \mathcal{O}(\sqrt{\Delta t_n}), \quad d_n - 1 = \mathcal{O}(\sqrt{\Delta t_n})$$

Since $\ln u_n = \mathcal{O}(\sqrt{\Delta t_n})$, $\ln d_n = \mathcal{O}(\sqrt{\Delta t_n})$ this proves: $\mathbf{p}_n = \mathcal{O}(\Delta t_n^2)$.

Proof of Proposition 3

Since Tian has explicitly fixed the first three moments: $\bar{\mathbf{m}}_n^1 = \bar{\mathbf{m}}_n^2 = \bar{\mathbf{m}}_n^3 = 0$, we only need to verify $\mathbf{p}_n = \mathcal{O}(\frac{1}{n^2})$.

From Abramowitz and Stegun (1968) we know the following

$$\forall 0 < z \leq 0.5829 : |\ln(1 - z)| < \frac{3}{2}z \Rightarrow \forall 0.4172 \leq z < 1 : |\ln z| \leq \frac{3}{2}|z - 1|$$

$$\forall z > 0 : \ln(1 + z) < z \Rightarrow \forall z > 1 : |\ln z| < \frac{3}{2}(z - 1)$$

This is fulfilled for sufficiently high n with $0.4172 < d_n < 1 < u_n$ thus yielding $(\ln u_n)^2 \leq \frac{9}{4}(u_n - 1)^2$, $(\ln d_n)^2 \leq \frac{9}{4}(d_n - 1)^2$.

Therefore

$$\begin{aligned} \mathbf{p}_n &= \bar{E}[(\ln \bar{R}_{k+1})(\bar{R}_{k+1} - 1)^3] \\ &\leq \bar{E}[(\ln \bar{R}_{k+1})^2]^{\frac{1}{2}} \bar{E}[(\bar{R}_{k+1} - 1)^3]^{\frac{1}{2}} \text{ by Hölder's Inequality} \\ &\leq \frac{9}{4} \bar{E}[(\bar{R}_{k+1} - 1)^2]^{\frac{1}{2}} \bar{E}[(\bar{R}_{k+1} - 1)^3]^{\frac{1}{2}} \end{aligned}$$

Thus it is sufficient to prove: $u_n - 1 = \mathcal{O}(\sqrt{\Delta t_n})$, $d_n - 1 = \mathcal{O}(\sqrt{\Delta t_n})$

Forming a series expansion

$$v_n^2 + 2v_n - 3 = 1 + \mathcal{O}(\Delta t_n) + 2 + \mathcal{O}(\Delta t_n) - 3 = \mathcal{O}(\Delta t_n)$$

and

$$\begin{aligned} u_n - 1 &= \frac{r_n v_n}{2} (v_n + 1 + (v_n^2 + 2v_n - 3)^{1/2}) - 1 \\ &= \frac{r_n v_n}{2} (v_n + 1 + \mathcal{O}(\sqrt{\Delta t_n})) - 1 = \frac{r_n v_n^2}{2} + \frac{r_n v_n}{2} + \mathcal{O}(\sqrt{\Delta t_n}) - 1 \\ &= \frac{1 + \mathcal{O}(\sqrt{\Delta t_n})}{2} + \frac{1 + \mathcal{O}(\sqrt{\Delta t_n})}{2} + \mathcal{O}(\sqrt{\Delta t_n}) - 1 \\ &= \mathcal{O}(\sqrt{\Delta t_n}) \end{aligned}$$

In the same way one proves: $d_n - 1 = \mathcal{O}(\sqrt{\Delta t_n})$.